SHUBH TAMRAKAR

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PROFESSIONAL SUMMARY

B.Tech Computer Science (Artificial Intelligence) student seeking a software development or AI/ML internship. Built a Raspberry Pi biometric voting prototype and a real-time chat platform using React and Firebase, and currently developing a Scikit-learn tool to analyze model failures. Completed IBM's Building AI Agents and Agentic Workflows specialization (LangGraph, CrewAI, BeeAI, AG2) and placed third runner-up among 1900+ registrations at a national hackathon. Skilled in Python, Java, JavaScript, and Git.

EDUCATION

Pranveer Singh Institute of Technology (PSIT) Sept. 2024 – Aug. 2028 (Expected)
Bachelor of Technology, Computer Science and Engineering (Artificial Intelligence) — Kanpur, India
Additional: Class XII (Mar. 2024) and Class X (Apr. 2022), Christukula Mission Higher Secondary School, Satna, M.P.

ACHIEVEMENTS

Third Runner-Up — AHGV Buildverse 2026 (Paytm AI Hackathon) 2026
- Selected among the top 8 teams out of 1900+ registrations on Unstop, and secured the third runner-up position.

Grand Finalist — TechEXPO (PSIT College Fest) 2026
- Selected through the qualifying rounds to compete in the grand final of the college fest TechEXPO.

PROJECTS

BioVote — Secure Voting System Using Biometric Verification https://github.com/Happybro2005/BIO-VOTE
Raspberry Pi • Python • Fingerprint Sensor • VVPAT
- Built a Raspberry Pi–based voting machine prototype that authenticates voters through fingerprint scanning with Aadhaar-based verification.
- Displays the voter's photo and name for verification, then lets them cast a vote for a candidate.
- Prevents duplicate and fake voting by rejecting repeat attempts, with votes stored in a local database and a VVPAT slip printed once per voter.
- Designed LAN-based synchronization across machines and real-time turnout tracking with audit-friendly result records.

NovaTalk — Real-Time Social Communication Platform https://github.com/Happybro2005/NovaTalk
React • Vite • Firebase (Auth, Firestore, Realtime Database, Storage)
- Developed a real-time chat platform with anonymous stranger matching, friend networking, communities, and live messaging.
- Implemented typing indicators, read receipts, presence tracking, and image and voice-note sharing using Firebase real-time services.
- Built an admin Mission Control panel for user management, report moderation, analytics, and audit logs with role-based access control.
- Secured the app with Firebase Authentication, protected routes, and Firestore, Realtime Database, and Storage security rules.

ML Model Failure Analyzer (In Progress)
Python • Scikit-learn
- Developing a machine learning diagnostic system to analyze model predictions and identify patterns behind incorrect classifications.
- Implementing data splitting, baseline classifier training, and prediction generation using Scikit-learn.
- Generating class predictions and confidence scores to investigate high-confidence errors and misclassifications.
- Preparing structured prediction datasets (actual labels, predicted labels, confidence scores, input features) for downstream error analysis.

ETBLM — Responsive E-commerce Website https://github.com/ShubhTamrakar/ETBLM
HTML • CSS • JavaScript
- Developed a responsive floral e-commerce website with product pages, category browsing, and interactive navigation.
- Optimized for desktop and mobile using responsive web design principles, with a clean UI and maintainable code structure.

CERTIFICATIONS

Building AI Agents and Agentic Workflows Specialization — IBM, Oct. 2026 Verify specialization certificate
- Built LangGraph agents with memory, iteration, conditional logic, and Agentic RAG; developed self-improving and multi-agent systems with orchestration.
- Created structured, modular workflows with CrewAI and applied IBM BeeAI and AG2 (AutoGen) to build conversation-driven, tool-using agent applications.
- Includes three IBM courses, each listed with its own certificate below.

Fundamentals of Building AI Agents — IBM (Coursera), Oct. 2026 https://coursera.org/verify/9LND0EWKSUXF
Agentic AI with LangChain and LangGraph — IBM (Coursera), Oct. 2026 https://coursera.org/verify/82LX11ELTTM5
Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI — IBM (Coursera), Oct. 2026 https://coursera.org/verify/NjZVET2R3XXL

TECHNICAL SKILLS

Programming Languages: Python, Java, C, JavaScript, HTML, CSS
Frameworks: React, Vite, Scikit-learn, NumPy, Pandas
AI/Agents: LangGraph, CrewAI, AutoGen (AG2), BeeAI, Agentic RAG
Databases: Firebase Firestore, Firebase Realtime Database
Tools/Platform: Git, GitHub, VS Code, Firebase, Raspberry Pi
Core Concepts: Object-Oriented Programming (OOP), Data Structures, Responsive Web Design, Problem Solving